

TRIASHA SARKAR

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Machine Learning Engineer and Georgia Tech aerospace MS graduate who builds dependable systems from data and experiments through deployment. Experience spans retrieval and ranking, scientific ML, evaluation, distributed training, and high-throughput serving.

EDUCATION

MS, Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA | Aug 2026

CGPA: 3.7/4.0 | Entered PhD program; transitioned to MS and completed the degree in Aug 2026

MSc, Aerospace Vehicle Design | Cranfield University, United Kingdom | Feb 2023

CGPA: 3.6/4.0 | First Class Honours | Thesis on geometry optimization under FAA constraints

B.Tech, Aerospace Engineering | SRM Institute of Science and Technology, India | Jun 2021

CGPA: 3.8/4.0 | First Class Honours

TECHNICAL SKILLS

ML and GenAI: PyTorch, scikit-learn, Transformers, PEFT/LoRA, RAG, BM25, dense retrieval, reranking, recommender systems, uncertainty estimation

Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, PostgreSQL/pgvector, Prometheus, OpenTelemetry, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, NetworkX, OSMnx, statistical inference, design of experiments, surrogate modeling

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris | Georgia Tech ASDL | May 2025 - Aug 2026

- Took simulation, safety, and sustainability questions from data review and model design through tested analysis, with reproducible evidence for faculty and sponsors.
- Checked assumptions with domain experts, documented limitations, and delivered models and workflows that others could rerun and use.

Machine Learning Engineer | Rolls-Royce | Jul 2023 - Apr 2025

- Built Python workflows for diagnostics, anomaly detection, predictive maintenance, and mixed-frequency time series from multivariate engine data.
- Turned certification requirements into traceable analyses and model checks, reviewing each result with lifecycle engineers before use.

Data Science Intern | Rolls-Royce DataLabs | May - Jul 2021

- Cleaned aircraft-engine sensor data, designed features, and compared predictive and anomaly-detection models to identify recurring failure patterns.

TECHNICAL PROJECTS

AeroRAG-X | Retrieval, post-training, and ML systems | 2025 - Present

- Built hybrid retrieval, reranking, pgvector search, evidence gating, and citation controls over 3,233 source-preserving NASA NTRS chunks.
- On 888 QASPER questions with human evidence spans, a frozen TF-IDF baseline reached 76.24% evidence recall@10. This is retrieval evidence on NLP papers, not NASA answer generation.

IntegrityBench | Moderation evaluation and release controls | Aug 2026 - Present

- Trained a Civil Comments candidate with validation-only safety thresholds, then tested it without retraining on 2,802 human-annotated ToxicChat prompts.
- False acceptance rose from 1.84% on Civil Comments to 59.32% on ToxicChat, so the release remains blocked rather than presented as a general moderation model.

NewsLens | Recommendation and real-time search | Jul - Aug 2026

- Reached NDCG@10 of 0.366 with leakage-safe MIND evaluation; the one-time holdout interval included zero.
- Built a Go, Kafka, PostgreSQL, and FastAPI article path; a 500-event local run measured 44 ms publish p99, 79 ms freshness p95, and 5.7 s partition recovery.

EdgeGenBench | Real-flight ML and on-device inference | Aug 2026 - Present

- A NASA DASHlink anomaly model reached 0.7380 macro F1 on 17,780 aircraft-disjoint approaches and remained blocked by release gates.
- ONNX prediction consistency stayed above 99.55% under tested sensor corruptions; generated aircraft-design data is labeled separately as deployment evidence.

AeroSynth-Eval | Multimodal AI evaluation | Aug 2026 - Present

- On public AGDD aircraft images, mixed training raised mean macro F1 from 0.3881 to 0.4419 but reduced crack recall from 0.4000 to 0.3167.
- Materialized 1,735 GenAI-Bench human preference votes into prompt-grouped train, validation, and held-out partitions for general image-evaluator development, not aircraft-inspection validation.

Surrogate Model Learning | Real-data reliability studies | Aug 2026 - Present

- Built grouped-split UCI airfoil and building-load experiments with seed sensitivity, extrapolation checks, and conformal intervals.
- A normalized conformal diagnostic reached 87.93% and 87.07% coverage against a 90% target; it remains a retrospective finding pending untouched-data confirmation.

Atlanta Mobility Resilience Digital Twin | 2026 - Present

- Built an OSMnx and NetworkX road-disruption simulator, then added 50 Census tract origins from 2024 ACS estimates representing an estimated 216,659 residents.
- Retained ACS margins of error and source hashes. Essential destinations and observed traffic calibration remain open.

Silent Failure Detection in Rocket-Motor Simulations | Georgia Tech ASDL | May - Jul 2026

- Audited 15,120 simulations, built a leakage-safe classifier with 96.8% unseen-geometry accuracy, and traced silent failures to an uncapped convergence loop.

HERO | Source-aware retrieval for airline safety | Georgia Tech ASDL | 2025 - 2026

- Worked on retrieval and evaluation methods that keep generated safety answers tied to technical sources analysts can review.

GREEN TEA and Project EAGLE | Sustainable aviation modeling | Sep 2025 - Apr 2026

- Built surrogate, uncertainty, demand, and life-cycle models for alternative-fuel and transport studies; the GREEN TEA model remains in sponsor use.

Equity Backtest | Walk-forward signal evaluation | Jul 2026

- Built an expanding-window backtest with transaction costs and retained every specification tested; the apparent momentum edge disappeared after costs and missed the preregistered significance threshold.

LEADERSHIP

Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech | 2025 - 2026